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Sources and Mechanism of Lithium Enrichment in Oilfield Brines

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Abstract

The increasing demand for lithium (Li), driven by the transition to sustainable energy solutions, necessitates a deeper understanding of the sources and mechanisms of Li enrichment in unconventional reservoirs, such as oilfield brines. This study presents a conceptual model that integrates geological, hydrogeological, and geochemical processes to explain the formation of Li-rich oilfield brines. Our model identifies primary Li sources, including Li-bearing igneous rocks and sedimentary deposits, and highlights the critical role of tectonic activity, fault systems, and hydrothermal fluid migration in transporting Li from deep-seated origins into oilfield brines. We also emphasize the importance of hydrologically closed basins, arid climates, and evaporative processes in concentrating Li and other dissolved solids. The model provides a framework for understanding the complex interplay of geological and hydrogeological factors that control Li enrichment in oilfield brines, and have implications for the global exploration of Li resources. By applying this conceptual model, explorers and researchers can identify prospective areas for Li discovery and development, ultimately contributing to a more sustainable and secure Li supply chain.

Introduction

Lithium (Li), a crucial element in the production of high-energy batteries, has emerged as a vital component in the global quest for sustainable energy solutions. As the world accelerates its transition to a low-carbon economy, the imperative to reduce internal combustion engine use in transportation has become more pressing, driven by the soaring demand for electric vehicles, renewable energy storage systems, and portable electronic devices. Consequently, the global demand for Li is experiencing an unprecedented surge, with forecasts indicating a continued upward trend, as Li and other essential battery materials play an increasingly critical role in enabling the widespread adoption of clean energy technologies and propelling the world towards a carbon-neutral future (Parvizi et al., 2025).

Traditionally, Li has been sourced from conventional hard rock mining operations, primarily from spodumene-bearing pegmatites (Fortier et al., 2019). However, these methods are often associated with significant environmental impacts, including habitat disruption, high energy consumption, and substantial waste generation (Kaunda, 2020). In contrast, Li extraction from Salar brines, particularly in South America, has emerged as a more efficient and cost-effective alternative (Bustos-Gallardo et al., 2021). While Salar brine operations offer advantages such as lower capital expenditure and easier extraction processes, they are

not without their challenges, including long evaporation times, water resource depletion, and environmental concerns related to brine disposal (Marazuela et al., 2020).

Recognizing the limitations of conventional and Salar-based Li extraction, there is a growing interest in exploring alternative brine resources, such as oilfield brines and geothermal fluids. These brines can contain elevated Li concentrations and offer several potential advantages over rock mining, including reduced environmental footprint, utilization of existing infrastructure, and the potential for co-production with other valuable minerals (Stringfellow and Dobson, 2021). One of the key challenges in exploring for Li in oilfield brines is understanding the origin of the Li. Li is a highly reactive element that can be introduced into oilfield brines through a variety of mechanisms, including the interaction of formation waters with volcanic rocks, the dissolution of Li-rich minerals, and the migration of Li-rich fluids from deeper reservoirs (Munk et al., 2016).

The Arabian Peninsula is a region of significant interest for Li exploration due to its unique geological setting. The peninsula has a long history of oil and gas production, and the resulting oilfield brines might contain significant amounts of Li. However, the exploration for Li in oilfield brines is still in its early stages, and there is a need for further research to understand the geological processes that control Li concentration in these systems. To understand the origin of Li in oilfield brines, a multiple line of evidence approach was applied in this study to evaluate the Arabian Peninsula as a potential source of Li in oilfield brines.

Geological Setting

The Arabian Platform is a geological region that covers a significant portion of the Arabian Peninsula, including Saudi Arabia, Yemen, Oman, United Arab Emirates, Qatar, Bahrain, and Kuwait. The platform is a stable continental region that has been shaped by a combination of tectonic, volcanic, and sedimentary processes over millions of years (Stern et al., 2010).

Historically, the Arabian Platform has a long and complex geological history that dates back to the Precambrian era, over 2.5 billion years ago. During this time, the platform was part of the supercontinent of Gondwana, which included Africa, South America, Australia, and Antarctica (Torsvik et al., 2013). The platform was formed as a result of the collision between the Arabian-Nubian Shield and the Eurasian Plate, which led to the formation of a series of mountain ranges, including the Arabian Shield (Stern et al., 2010). During the Phanerozoic era, the Arabian Platform was subjected to a series of tectonic events, including rifting, faulting, and folding, which led to the formation of several sedimentary basins (Nairn and Alsharhan, 1997). These basins, including the Arabian Basin, the Rub' al Khali Basin, and the Gulf Basin, are filled with a thick sequence of sedimentary rocks, including sandstones, limestones, and shales, which were deposited in a variety of environments, including fluvial, marine, and eolian (Nairn and Alsharhan, 1997). Furthermore, the Arabian Platform is characterized by a series of structural features, including faults, folds, and fractures, which were formed as a result of the tectonic events that have shaped the region. The platform is bounded by several major faults, including the Arabian Fault, the Red Sea Rift, and the Gulf of Oman Fault, which have played a significant role in shaping the region's geology (Al-Husseini, 2000).

Hydrogeologically, the Arabian Platform is characterized by a complex hydrogeological system, with several major aquifers and aquifer systems that play a crucial role in the region's water resources (Saeed et al., 2021). The Arabian Platform is also characterized by the presence of formation water, which is water that is trapped in the pores of rocks and sediments. The formation water in the region is highly saline, with total dissolved solids (TDS) ranging from 10,000 to more than 300,000 milligrams per liter (Saeed et al., 2021). The formation water is also highly variable in terms of its chemical composition, with different aquifers and formations having distinct chemical signatures.

Source of Lithium in Oilfield Brines

The presence of Li in oilfield brines is often attributed to the dissolution of primary Li-bearing igneous rocks, which are typically of volcanic origin (Bradley et al., 2013; Munk et al., 2016). These rocks, such as pegmatites and felsic volcanic lithologies like rhyolite and trachyte, contain Li-rich minerals like spodumene, lepidolite, and petalite. Notably, Li concentrations are highest in felsic rocks, which are characterized by their high silica content and low iron and magnesium content (Gourcerol et al., 2019). In contrast, intermediate rocks have lower Li concentrations, while mafic and ultramafic rocks are generally Li-poor due to their low silica content and high iron and magnesium content (Gourcerol et al., 2019). The weathering and erosion of these Li-rich rocks release Li ions, which are then transported by water, often in the form of Li-containing brines (Benson et al., 2017). The interaction of basin-fill sediments with geothermal and hydrothermal fluids also plays a crucial role in the development of Li-rich brines (Coffey et al., 2021).

In the context of the Arabian Peninsula, the region's complex geodynamic evolution has introduced a wide range of Li sources. Foremost among these, the ancient Precambrian Arabian Shield, which covers a significant portion of the peninsula, is composed of volcanic and metamorphic rocks that could potentially host Li-rich minerals. Indeed, the Shield is notably rich in rocks suitable for economic concentrations of Li, including pegmatites and granites (Drysdall and Douch, 1986; Küster, 2009). Pegmatite, in particular, could be a primary Li source due to its characteristic high content of alkaline elements. Beyond these older sources, the region has also experienced widespread magmatic activity since the late Permian. The break-up of Pangaea-Gondwana and the subsequent formation of the Tethys Ocean led to major phases of magmatism, including mantle plume activity and localized magmatic activity within intra-plate rifts (Sharland et al., 2004). This younger magmatic activity has resulted in the formation of a diverse range of igneous rocks, including those rich in Li.

Understanding the regional correlations between magmatic activity and the major tectonic events (including basin subsidence, basement uplift, and intraplate deformation) that have shaped the Arabian Peninsula is pivotal for deciphering its geodynamic evolution. This comprehensive understanding is crucial for identifying potential Li sources. Magmatic activity directly influences the lithosphere's thermal evolution, a key factor in the formation and concentration of Li-rich brines. Therefore, a thorough grasp of the Arabian Peninsula's intricate magmatic history is indispensable for guiding Li resource exploration. By accurately pinpointing areas with both abundant felsic rocks and a documented history of magmatic activity, effective strategies for discovering and extracting this valuable resource can be developed.

Transport of Lithium from Source to Oilfield Brines

The journey of Li from its geological source to becoming a valuable component of sedimentary basin brines is a critical process in forming economic Li-rich deposits. Li can enter a basin through several key mechanisms, ranging from direct atmospheric deposition to deep-seated hydrothermal fluid migration. Understanding these diverse transport pathways is essential for pinpointing potential Li resources.

Surface and Near-Surface Transport Mechanisms

Direct Ashfall. Li can be transported and introduced into basins directly through ashfall, particularly from felsic volcanic eruptions. The effectiveness of this mechanism is highly dependent on factors such as the size of the ash particles, the explosiveness of the volcano, and atmospheric transport patterns. Generally, the more proximal a basin is to a felsic volcanic source, the higher the likelihood of increased Li contribution from volcanic ashfall. Over geological timescales, extensive ashfall events can provide a significant initial input of Li to basin sediments.

Weathering and Erosion. The most common pathway involves the weathering and erosion of surficial Li-bearing rocks and their subsequent transport by water (Benson et al., 2017; Stephenson, 2023). As Li-rich volcanic rocks, such as pegmatites and granites, break down, Li is released and carried into basins via surface drainage networks (Bradley et al., 2013). The erosion of ancient volcanic landscapes can continuously feed Li into a basin system, where meteoric waters then play a crucial role in its initial concentration.

Li Release from Clay Deposits. An important secondary mechanism involves Li's interaction with clay minerals. Li leached from rhyolitic lavas and volcanic ash by meteoric waters can become structurally bound within clay deposits. While initially immobile, this Li can later be released from the clay matrix by subsequent hydrothermal fluids. This mechanism highlights how Li, once incorporated into sedimentary structures, can be remobilized under the right thermal and fluid conditions.

Deep-Seated Hydrothermal Transport Mechanisms

Beyond surface processes, tectonic activity plays a paramount role by creating pathways for Li migration from deeper realms, often through hydrothermal fluids. These fluids can be locally introduced due to the subsidence of geological formations or via extensive fault and fracture networks.

Subsidence-Driven Hydrothermal Activity. The subsidence, or sinking of land, within the Arabian Peninsula, particularly across the Arabian Platform, has profoundly shaped its geological formations and significantly contributed to hydrothermal activity. This process is driven by tectonic activity, often augmented by the unique geological properties of the region, such as the presence of evaporite deposits (like salt) and active groundwater circulation. The subsidence of a geological formation due to tectonic activity has a significant impact on the hydrothermal characteristics of the groundwater trapped within that subsiding formation. This impact primarily revolves around:

- **Increased Temperature:** As formations subside, they are buried deeper, exposing trapped groundwater to higher temperatures due to the geothermal gradient. This elevated temperature accelerates water-rock interactions.
- **Increased Pressure:** The weight of accumulating overlying sediments increases both lithostatic and hydrostatic pressures on the trapped fluids, which can lead to over-pressuring and drive fluid movement.
- **Modified Fluid Flow Pathways:** Compaction-driven flow from deeper, compacting sediments can push fluids upwards into more permeable units. The ongoing tectonic activity associated with subsidence can also generate new fractures or reactivate existing ones, creating or enhancing fluid migration pathways.
- **Enhanced Fluid Chemistry:** Elevated temperatures and pressures at depth significantly enhance the rate and extent of water-rock interactions, allowing groundwater to dissolve and become enriched in various elements, including Li.

Reconstructions of both the sedimentary depositional history and geodynamic evolution of the Arabian Platform corroborate this. Subsidence modeling of the Permian to Jurassic succession of the Arabian Platform indicates several major tectonic events during the Late Permian, Early Triassic, Norian, Middle Toarcian, and Dogger (Le Nindre et al., 2003). This history of episodic subsidence has played a crucial role in creating environments conducive to hydrothermal activity and the deep-seated maturation of brines.

Fault-Driven Hydrothermal Migration. Beyond subsidence, active faults and associated fault systems provide vital, direct pathways for Li migration from deeper realms (Li et al., 2019). Active faults, especially where crossed by drainage networks, act as conduits for Li-rich magmatic and hydrothermal fluids to ascend from depth and enter the basin. These fluids can dramatically increase the Li concentration, particularly in regions with significant tectonic activity. The combined effect of these transport mechanisms is essential,

as they facilitate the introduction of Li into a basin, where it can then be further concentrated through evaporation and other geochemical processes.

Faults are paramount in this transport, enabling the upward migration of hot hydrothermal fluids from deep sources into shallower reservoirs. These deep-seated fluids can either bring Li directly from their volcanic or magmatic origins, or, upon encountering shallower rocks already rich in Li, they can induce water-rock interactions. These interactions lead to the dissolution and mobilization of Li from the rock matrix, effectively "flushing" it into the migrating fluid. Compelling evidence for such rapid, deep-seated fluid migration through faults is observed across the Arabian Peninsula. For instance, fluid inclusion data from the northern and eastern Arabian Peninsula strongly indicates that hydrothermal fluids actively migrated up-fault from deeper sources during the Zagros Orogeny. This significant tectonic event caused the release of pressurized hydrothermal fluids as faults ruptured (Hollis, 2011). Further evidence comes from northern Oman, where an evaluation of fluid-flow systems shows that hydrothermal brines moved up-fault during Upper Cretaceous and Cenozoic structural events, originating from deeper sections or along a basal décollement. These observations highlight the region's long history of deep fluid circulation driven by tectonics. Specific examples from key oilfields provide concrete support for this proposed mechanism.

- Southwest Iran: Goff (2005) documented that Cretaceous formations received Upper Jurassic dolomitizing fluids that migrated up-fault through a thin Gotnia Formation. Jurassic reservoirs in this region display homogenization temperatures (TH) between 90°C and 120°C, further supporting deep, hot fluid flow.
- South Kuwait: Fluid inclusion data from Cretaceous and Jurassic reservoirs provide insights into the origin of hot fluids. The preservation of these high temperatures suggests rapid migration from depth through deep-seated fault systems. This implies that very hot fluids likely moved up-fault from near the basement into the Cretaceous section (Al-Hajeri et al., 2017).

These numerous documented instances of rapid, up-fault migration of hot hydrothermal fluids from deep sources throughout the Arabian Peninsula offer robust evidence for a crucial mechanism. They demonstrate how Li-rich brines can be effectively transported from their primary geological sources (e.g., magmatic bodies, or shallow rocks leached by these hot fluids) into shallower sedimentary basins, potentially contributing to the formation of Li-enriched oilfield brines. This understanding is key to pinpointing potential Li resources within such complex geological settings.

Accumulation of Lithium-Rich Brines

The successful formation of economically viable sedimentary Li deposits hinges critically on the presence of hydrologically closed basins (Bradley et al., 2013). These basins don't just act as traps; they function as natural evaporative engines (Stephenson, 2023). By restricting water outflow, they create ideal environments for intense evaporation, especially under arid climatic conditions. This continuous evaporative process progressively concentrates Li along with other dissolved solids, ultimately leading to the genesis of highly Li-rich brines.

The Arabian Peninsula's inherently arid climate, characterized by exceptionally high evaporation rates and minimal precipitation, significantly boosts the potential for lithium accumulation within these unique hydrological settings. Crucially, this aridity isn't merely a modern phenomenon; the Arabian Peninsula has experienced prolonged periods of aridity and the formation of restricted water bodies (such as ancient lakes or seas) throughout its geological history (Böhme et al., 2021; Hallam, 1984; Mansour et al., 2024). These conditions were often driven by tectonic activity and shifting global climates. The widespread presence of evaporite strata, particularly those rich in halite and anhydrite, across the Arabian Peninsula serves as a definitive geological indicator of these past arid conditions and high evaporation rates within closed basins (Beydoun, 1986; Nairn and Alsharhan, 1997; Christian, 1997; Martin, 2001). These evaporites form

precisely when dissolved salts in water become so concentrated due to evaporation that they precipitate out of solution and accumulate as thick layers of rock. This deep geological history of aridity and restricted water bodies thus highlights the peninsula's long-standing potential as a region for Li accumulation.

Furthermore, the widespread discovery of trapped paleo-seawater within the region's vast hydrocarbon reservoirs serves as compelling evidence that the precise conditions for substantial lithium accumulation have been met repeatedly throughout its geological past. For instance, detailed chemical and isotopic fingerprinting techniques have robustly confirmed an evaporative seawater origin for hypersaline waters in the Cretaceous Formation of the Rub' al Khali basin (Saeed et al., 2021) firmly indicates that the primary source of dissolved solutes in these brines is directly linked to the entrapment of evaporated paleo-seawater (connate water) in ancient nearshore and lagoonal environments at the time of deposition. These evidentially evaporated brines are a direct indication of progressive evaporation within hydrologically closed basin conditions. Consequently, any Li-rich fluids introduced into these confined systems would undergo significant concentration and accumulation, forming enriched Li brines.

Conclusion

The escalating global demand for lithium, driven by the imperative for sustainable energy solutions, necessitates the exploration of alternative, environmentally sound extraction methods. This manuscript employed a multi-faceted approach to evaluate the potential of the Arabian Peninsula's oilfield brines as a significant lithium resource, moving beyond conventional hard rock and Salar brine sources.

Our comprehensive analysis of the Arabian Platform's unique geological setting reveals a region with a long and complex tectonic and magmatic history, characterized by extensive sedimentary basins and highly saline formation waters. Crucially, the presence of primary Li sources is well-established; the ancient Precambrian Arabian Shield, rich in pegmatites and granites, alongside widespread younger magmatic activity associated with Pangaea-Gondwana break-up and intra-plate rifting, provides ample felsic igneous rocks capable of hosting Li-rich minerals.

Furthermore, the study elucidated the critical mechanisms for Li transport from these sources into oilfield brines. Tectonic activity and associated fault systems are paramount, acting as conduits for the upward migration of hot, Li-rich hydrothermal fluids from deep-seated origins. Compelling evidence from fluid inclusion data across the Arabian Peninsula, including the Southwest Iran, and South Kuwait, robustly supports rapid, up-fault fluid migration, demonstrating an effective pathway for Li-rich brines to enter shallower sedimentary basins.

Finally, the accumulation of Li-rich brines is strongly supported by the region's hydrogeological characteristics. The Arabian Peninsula's inherently arid climate, coupled with a deep geological history of hydrologically closed basins, has created ideal conditions for intense evaporation and the progressive concentration of dissolved solids, including Li. The widespread occurrence of evaporite strata and the robust evidence of trapped paleo-seawater within hydrocarbon reservoirs underscore the long-standing potential for significant Li accumulation.

In conclusion, the convergence of primary Li sources, efficient transport mechanisms via fault systems, and favorable conditions for brine accumulation strongly indicates that the Arabian Peninsula's oilfield brines represent a promising frontier for Li exploration (Figure 1). This integrated understanding of the region's geological and hydrogeological evolution provides a robust framework for guiding future exploration efforts and contributes significantly to the global quest for sustainable Li resources to power a carbon-neutral future.

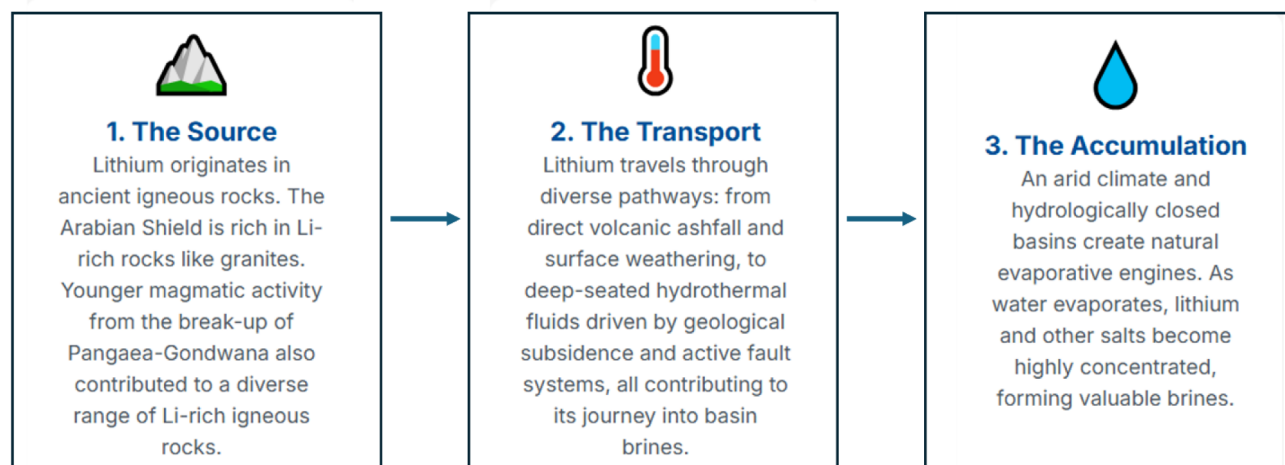


Figure 1—The infographic highlights the significant potential of the Arabian Peninsula's oilfield brines as a new source for lithium, driven by the region's unique geology, efficient transport mechanisms, and ideal conditions for accumulation.

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